MIL-S-19500/380A(EL)
17 September 1968

SUPERSEDING
MIL-S-19500/380(EL)
8 March 1968

MILITARY SPECIFICATION

SEMICONDUCTOR DEVICE, TRANSISTOR, NPN, SILICON, POWER
TYPES 2N4865, 2N5250, 2N5251 and TX2N4865, TX2N5250, TX2N5251

1. SCOPE

1.1 Scope.- This specification covers the detail requirements for silicon, NPN, power transistors for particular use in power-switching, electronic-circuit applications. (See 3.3 and 6.2 herein.) The prefix "TX" identifies devices meeting the special process-conditioning, testing, and screening requirements in 4.4 herein, and the stringent quality-control (LTPD) requirements in Tables I, II, and III herein.

1.2 Outline and dimensions.- See Figure 1 herein.

1.3 Maximum ratings.-

	P_T 1/			θ_{J-C}	V_{CBO}	V_{CEO}	V_{EBO}	I_C 2/	$T_{(oper)}$	T_{stg}
	at: T_C= +25°C	at: T_C= +50°C	at: T_C= +100°C							
	W	W	W	°C/W	Vdc	Vdc	Vdc	Adc	°C	°C
2N4865	350	300	200	0.5	100	80	10	50	3/	4/
2N5250	350	300	200	0.5	125	100	10	50	3/	4/
2N5251	350	300	200	0.5	180	150	10	50	3/	4/

1/ Linear derating factor = 2.0 W/°C up to and including T_C = +200°C.
2/ Pulsed (see 4.3.1 herein) = 90A.
3/ Temperature range = -55° to +200°C.
4/ Temperature range = -65° to +200°C.

FSC-5961

Source: https://assist.dla.mil -- Downloaded: 202610-18T15:36Z
Check the source to verify that this is the current version before use.

1.4 Particular electrical characteristics.- (At T_C= +25°C):

	h_{FE} 1/								$V_{BE(sat)}$ 1/			
	at: I_C=90Adc V_{CE}=5Vdc		at: I_C=70Adc V_{CE}=5Vdc		at: I_C=40Adc V_{CE}=5Vdc		at: I_C=20Adc V_{CE}=5Vdc		at: I_C=70Adc I_B=10Adc		at: I_C=40Adc I_B=4Adc	
	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max
	---	---	---	---	---	---	---	---	Vdc	Vdc	Vdc	Vdc
2N4865	5.0	---	10	40	15	50	30	120	---	2.5	---	1.8
2N5250	5.0	---	10	40	15	50	30	120	---	2.5	---	1.8
2N5251	5.0	---	10	40	15	50	30	120	---	2.5	---	1.8

1/ Pulsed: see 4.3.1 herein.

	$V_{CE(sat)}$ 1/				h_{fe}		t_{on}	t_s	t_f	t_{on}	t_s	t_f
	at: I_C=70Adc I_B=10Adc		at: I_C=40Adc I_B=4Adc		at: f=10MHz V_{CE}=10Vdc I_C=5Adc		at: I_C=70Adc (See Fig. 3A herein)			at: I_C=40Adc (See Fig. 3B herein)		
	Min	Max	Min	Max	Min	Max	Max	Max	Max	Max	Max	Max
	Vdc	Vdc	Vdc	Vdc	---	---	usec	usec	usec	usec	usec	usec
2N4865	---	2.0	---	1.0	1.0	7.0	2.0	1.5	0.5	2.0	1.5	0.5
2N5250	---	2.0	---	1.0	1.0	7.0	2.0	1.5	0.5	2.0	1.5	0.5
2N5251	---	2.3	---	1.35	1.0	7.0	2.0	1.5	0.5	2.0	1.5	0.5

1/ Pulsed: see 4.3.1 herein.

Source: https://assist.dla.mil -- Downloaded: 202610-18T15:36Z
Check the source to verify that this is the current version before use.

2. APPLICABLE DOCUMENTS

2.1 The following documents, of the issue in effect on date of invitation for bids or request for proposal, form a part of this specification to the extent specified herein:

SPECIFICATIONS

MILITARY

MIL-S-19500	Semiconductor Devices, General Specification for

STANDARDS

MILITARY

MIL-STD-202	Test Methods for Electronic and Electrical Component Parts
MIL-STD-750	Test Methods for Semiconductor Devices

(Copies of specifications, standards, drawings, and publications required by contractors in connection with specific procurement functions should be obtained from the procuring activity or as directed by the contracting officer. Both the title and number or symbol should be stipulated when requesting copies.)

2.2 Other publications.- The following document forms a part of the specification to the extent specified herein. Unless otherwise indicated, the issue in effect on date of invitation for bids or request for proposal shall apply.

NATIONAL BUREAU OF STANDARDS

Handbook H28 Screw-Thread Standards for Federal Services.

(Application for copies should be addressed to the Superintendent of Documents, Government Printing Office, Washington, D.C. 20402.)

3. REQUIREMENTS

3.1 Requirements.- Requirements for the transistors shall be in accordance with Specification MIL-S-19500, and as otherwise specified herein.

Source: https://assist.dla.mil -- Downloaded: 202640-18T15:36Z
Check the source to verify that this is the current version before use.

3.2 Design and construction.- The transistor shall be of the design, construction, and physical dimensions specified on Figure 1.

3.2.1 Terminal arrangement.- The terminal arrangement on the transistor shall be as indicated in Figure 1.

3.2.2 Operating position.- The transistor shall be capable of proper operation in any position.

3.3 Performance characteristics.- The transistor performance characteristics shall be as specified in Tables I, II, and III herein. Except where specifically differentiated for respective transistor types (see 1.3, 1.4, and Tables I, II, and III herein), the performance requirements, including characteristics, ratings, and test conditions, apply equally to all transistor types covered herein. (See 3.3.1 below.)

3.3.1 "TX" special requirements.- Specified "TX" device-types shall, in addition to all other performance requirements herein, be process-conditioned, tested, and screened in accordance with 4.4 herein (see 3.4.1 herein). (At option of the manufacturer, the non-TX device types may be subjected to the same process-conditioning and testing.)

3.4 Marking.- Except as otherwise specified herein, marking shall be in accordance with Specification MIL-S-19500. The "manufacturer's identification" and "country of origin" may, at option of the manufacturer, be omitted from being marked directly on the semiconductor device covered herein.

3.4.1 "TX" identification marking.- Marking of devices conforming to "TX" and all other applicable requirements herein (see 3.3) shall include the prefix "TX" immediately preceding the type designation.

4. QUALITY ASSURANCE PROVISIONS

4.1 General.- Except as otherwise specified herein, the responsibility for inspection, general procedures for acceptance, classification of inspection, and inspection conditions and methods of test shall be in accordance with Specification MIL-S-19500, Quality Assurance Provisions.

Source: https://assist.dla.mil -- Downloaded: 202610-18T15:36Z
Check the source to verify that this is the current version before use.

4.2 Qualification and acceptance inspection.- Qualification and Quality Conformance inspection shall be in accordance with Specification MIL-S-19500, Quality Assurance Provisions, and as otherwise specified herein. For Qualification inspection, the requirement for tightened LTPD for each subgroup, as required in Specification MIL-S-19500, Appendix C, paragraph 50.2 shall not apply. Groups A, B, and C inspection shall consist of the examinations and tests specified in Tables I, II and III, respectively, herein. Quality Conformance inspection shall include inspection of Preparation for Delivery (see 5.1 herein).

4.2.1 Discriminate qualification testing.- The non-TX device types shall be considered subject to Qualification tests. It shall be understood that procurement of the TX device types will be made from sources for the approved non-TX devices as listed in Qualified Products List (QPL)-19500. Upon request and validation by the qualifying activity, qualification may be extended (and appropriate QPL-listing effected) to TX devices manufactured by approved sources for the non-TX devices.

4.2.2 Special Group A criteria for Qualification inspection.- For Qualification inspection, only 8 failures will be permitted for all Group A tests combined. Hereto, where the manufacturer deems that the data in paragraph 4.4.2 of Specification MIL-S-19500 are invokable, notification shall be made to the Chief, Components and Material Standardization Branch, Procurement and Production Directorate, Fort Monmouth, New Jersey 07703, Attention: AMSEL-PP-EM-2.

4.2.3 Group B-Group C life test samples.- Samples of non-TX transistor types, only, that have been subjected to Group B, 340-hour life test may be continued on test for 1000 hours in order to satisfy Group C life test requirements. These samples shall be predesignated, and shall remain subjected to the Group C 1000-hour evaluation after they have passed the Group B, 340-hour acceptance criteria. The cumulative total of failures found during 340-hour test and during the subsequent interval up to 1000 hours shall be computed for 1000-hour acceptance criteria. (See 4.2.4 below). "TX" transistor types shall be subjected to 1000-hour life tests under group B inspection requirements, and the lambda (λ) established thereto in Table II shall apply.

4.2.4 Group C testing.- Group C tests shall be performed on a lot every 6 months. (See Table III herein.) The contractor shall, throughout the course of a contract or order, permit the Government representative to scrutinize all test data and findings covering manufacturer's test program on Group C characteristics and parameters for the product concerned. Upon determination by the Government inspector (in advance of Group C, 6-month, test results) that Group C parameters are not being adequately met, the Government inspector may require lot-by-lot inspection, normally for a minimum of 3 consecutive lots, to be performed for required Group C tests.

Source: https://assist.dla.mil -- Downloaded: 202610-18T15:36Z
Check the source to verify that this is the currentversion before use.

4.2.5 Disposition of sample units.- Sample units that have been subjected to Group B, Subgroup 2 and 5 tests shall not be delivered on the contract or order. Sample units that have been subjected to and have passed Group B, Subgroup 1, 3, 4, 6, 7, 8, 9, and 10, and all Group C tests (these tests to be considered non-destructive), may be delivered on the contract or order provided that, after Group B and C inspection is terminated, those sample units are subjected to and pass Group A inspection. Defective units from any sample group that may have passed group inspection shall not be delivered on the contract or order until the defect(s) has been remedied to the satisfaction of the Government.

4.2.6 Copies of lot-inspection data.- When specified in the contract or order, one copy of the Quality Conformance inspection data, including manufacturer's test data and findings, covering the device inspection lot being shipped shall accompany the device(s) concerned.

4.3 Particular examination and test requirements.-

4.3.1 "Pulsed"-conditions measurements.- Measurements required herein to be effected under pulsed conditions, shall be made in accordance with "Pulse Measurements" requirements in Section 4 of Standard MIL-STD-750.

4.3.2 Mechanical damage resulting from tests.- Except for intentionally deforming, mutilating, or dismembering mechanical-stress tests to which samples are subjected, there shall be no evidence of mechanical damage to any sample unit as a result of any of the Groups A, B, or C tests.

4.3.2.1 Durability of terminal lugs.- The lug ends of terminals shall remain intact on terminals throughout all physical, mechanical, and environmental tests herein.

4.3.3 Terminal Strength (Stud Torque) test.- Acceptance criteria for external threaded parts (see Fig. 1 herein) shall be in accordance with Handbook H-28.

4.3.4 Safe Operating Area (DC Operation) test.- Each transistor under test shall sustain the applicable test conditions and the following acceptance criteria shall apply:

(a) I_C (for each transistor) shall not vary more than ± 10% during the dc operation; and

Source: https://assist.dla.mil -- Downloaded: 2026-10-18T15:36Z
Check the source to verify that this is the current version before use.

(b) All other specified End-Point Test(s) limits shall not be exceeded, after the dc-operation test.

(c) Correlation note: *Satisfactory endurance of the transistors throughout Test 1, 2, 3 respectively (per Table II, Subgroup 6 herein) is directly associable with ascertainment of the "safe operating area" for the transistors as illustrated in the nomograph of Fig. 5 herein.*

4.4 Process conditioning, testing, and screening for "TX" devices.- All "TX" devices (100% of each lot) shall be subjected to the process conditioning, testing, and screening specified in 4.4.2 through 4.4.7.4 herein, in accordance with the procedural sequence established in Fig. 2 herein, prior to submission of the lot to Group A, B, C tests. The manufacturer shall permit the authorized Government representative to witness, concurrently with time of manufacturer's performance (of these tests), the process conditioning, testing, and screening on the devices. Those conditioning and screening tests normally performed by a manufacturer as standard-production tests, need not be repeated when these are predesignated and acceptable to the Government as being equal to or more severe than those specified herein.

4.4.1 Quality Assurance records and determinants.- The manufacturer shall maintain and make available for perusal and review at any time, during a period of 3 years minimum, to the authorized Government representative or other contracting customer concerned, complete lot-by-lot records for these devices. All devices that fail to meet the process-conditioning, testing, and screening requirements specified herein shall be removed from the respective lot, and the test-failure identity and quantity removed shall be recorded in the permanent lot history. (See 6.3, 6.4, and 6.5 herein.)

4.4.2 High-temperature storage.- The transistors shall be stored for at least 24 hours at a minimum temperature (T_{stg}) of +200°C.

4.4.3 Thermal shock (temperature cycling).- The transistors shall be subjected to temperature cycling (thermal shock) in accordance with Standard MIL-STD-750, Method 1051, Test Condition C, except that cycling duration shall be 10 continuous cycles and the exposure time at the temperature extremes shall be 15 minutes minimum.

Source: https://assist.dla.mil -- Downloaded: 202610-18T15:36Z
Check the source to verify that this is the current version before use.

4.4.4 Acceleration.- The transistors shall be subjected to the constant acceleration (centrifugal) test per Method 2006 in Standard MIL-STD-750, except that the following shall apply:

(a) Centrifugal force = 5,000G.

(b) Orientation = Y1, only.

(c) Hold-time of 1 minute shall not apply hereto.

4.4.5 Seal (fine-leak) test.- The transistors shall be subjected to the Seal test per Standard MIL-STD-202, Method 112, Test Condition C, procedure IIIa or IIIb (see 4.4.5.1 or 4.4.5.2 below), except that testing for gross leaks shall be in accordance with 4.4.5.3 below.

4.4.5.1 Conditions for procedure IIIa.- The sealed chamber in which the transistors are placed shall be pressurized with helium gas to 50 psig, minimum, for a minimum of 4 hours. At end of that period, the transistors shall be removed from the chamber and, within 30 minutes, be subjected to a helium leak-detection test. Those transistors showing a leakage rate of greater than 1×10^{-7} cc/sec of helium, when measured at a differential pressure of 1 atmosphere, shall be rejected and removed from the lot.

4.4.5.2 Conditions for procedure IIIb.- The devices shall be placed in an activation tank, pressurized with Krypton 85 tracer gas in a nitrogen solution, for sufficient time to detect a leak rate of 1×10^{-8} atmospheric cubic centimeters per second (atm cc/sec). Within four hours after subjection to this pressurization, the leak rate of the devices shall be determined on an attribute basis using the general equation shown below. Any device exhibiting a leak rate equal to or greater than 1×10^{-8} atm cc/sec shall be removed from the lot. The general equation for use with radioactive-gas leak test equipment is:

$$Q = \frac{R}{SKT(P_e^2 - P_i^2)}$$

Source: https://assist.dla.mil -- Downloaded: 2026-10-18T15:36Z
Check the source to verify that this is the current version before use.

where:

Q = leak rate in atm cc/sec.
R = net counting rate of tested part above background in cts/min.
S = specific activity of the test gas mixture in uCi/atm cc.
K = counting efficiency of the system for the given part in cts/min uCi.
P_e = pressure of test gas in activation tank during pressurization in atm abs.
P_i = pressure inside part under test in atm abs.
T = duration of pressurization in test gas mixture in seconds.

4.4.5.3 Gross leak test.- The transistors shall be immersed for a minimum period of 15 seconds in a suitable, non-corrosive liquid maintained at approximately 100°C, and shall be observed for emanating bubbles. Those transistors from which a bubble trace is observed shall be rejected and removed from the lot.

4.4.6 Reverse-bias burn-in.- The transistors shall be operated, for 48 ±4 hours, at T_A = +150°C, under the following test conditions:

2N4865: V_{CB} = 60 Vdc, I_E = 0
2N5250: V_{CB} = 80 Vdc, I_E = 0
2N5251: V_{CB} = 120 Vdc, I_E = 0

At end of the 48-hour period, the above-specified test voltages shall be maintained on the transistors until T_A = +30°±5°C is reached.

4.4.7 Burn-in conditioning, tests, and evaluation (screening).- To enable proper evaluation of the transistors conditioned and tested in accordance with the following requirements, each transistor and lot shall be handled or identified in such manner that results for each individual transistor and lot can be ascertained and recorded.

4.4.7.1 Pre burn-in measurements.- The transistors shall be tested for the characteristics listed in Table IV below, and the values found shall be recorded. Those transistors that fail to meet the specified limits shall be rejected and removed from the lot.

Source: https://assist.dla.mil -- Downloaded: 202610-18T15:36Z
Check the source to verify that this is the current version before use.

Table IV.- Pre burn-in measurements.

Test Method per MIL-STD-750	Examination or test 1/	Test Conditions	Symbol	Limits Min	Limits Max	Unit
3041	Collector-emitter cutoff current	Bias Cond. C V_{CE}=100Vdc V_{BE}= 0	I_{CES}	---	100	uAdc
3076	Forward-current transfer ratio	V_{CE}= 5Vdc I_{CE}= 70Adc Pulsed; 2/	h_{FE}	10	40	---

1/ See 3.3 herein. 2/ See 4.3.1 herein.

4.4.7.2 Burn-in conditioning.- The transistors shall be energized and operated for 168 hours minimum, under the following conditions:

T_C = +100°C
V_{CE} = 20Vdc
P_T = 200W

4.4.7.3 Post burn-in tests and measurements.- After burn-in conditioning, the transistors shall be permitted to return to room-ambient temperature thermal-equilibrium and retested, within a maximum total elapsed time of 96 hours, for the characteristics listed in Table IV above. The values now found for each individual transistor shall not exceed the following (respectively applicable to the pre burn-in measurements per Table IV above):

ΔI_{CES}= ±100% or 10 uAdc, whichever is greater
Δh_{FE}= ±20%

Source: https://assist.dla.mil -- Downloaded: 2026-10-18T15:36Z
Check the source to verify that this is the current version before use.

4.4.7.4 Burn-in conditioning evaluation (screening).- Those transistors whose measured characteristics exceed the delta (Δ) limits specified in 4.4.7.3, above, or the limits of Table IV, shall be rejected and removed from the lot. Where 10% or more of the transistors subjected to the Burn-in test are found to exceed those limits, the entire lot shall be considered non-acceptable as "TX" devices.

Source: https://assist.dla.mil -- Downloaded: 2026-10-18T15:36Z
Check the source to verify that this is the current version before use.

Table I.- Group A inspection.

Test Method per MIL-STD-750	Examination or test 1/	Conditions	LTPD Non-TX	LTPD TX	Symbol	Limits Min	Limits Max	Unit
	Subgroup 1		10	5				
2071	Visual and mechanical examination	---			---	---	---	---
	Subgroup 2		5	3				
3041	Collector-emitter cutoff current:	Bias Cond.C V_{BE}=0						
	2N4865	V_{CE}=100Vdc			I_{CES}	---	0.1	mAdc
	2N5250	V_{CE}=125Vdc			I_{CES}	---	0.1	mAdc
	2N5251	V_{CE}=180Vdc			I_{CES}	---	0.1	mAdc
3061	Emitter-base cutoff current	Bias Cond.D V_{EB} = 8Vdc			I_{EBO}	---	5.0	uAdc
3001	Collector-base breakdown voltage:	Bias Cond.D I_C=0.1 mAdc Pulsed; 2/						
	2N4865				BV_{CBO}	100	---	Vdc
	2N5250				BV_{CBO}	125	---	Vdc
	2N5251				BV_{CBO}	180	---	Vdc

Source: https://assist.dla.mil -- Downloaded: 202610-18T15:36Z
Check the source to verify that this is the current version before use.

Table I.- Group A inspection-(cont'd)

Test Method per MIL-STD-750	Examination or test 1/	Conditions	LTPD Non-TX	LTPD TX	Symbol	Limits Min	Limits Max	Unit
	Subgroup 2 (cont'd)							
3011	Collector-emitter breakdown voltage:	Bias Cond. D I_C= 200mAdc Pulsed; 2/						
	2N4865				BV_{CEO}	80	---	Vdc
	2N5250				BV_{CEO}	100	---	Vdc
	2N5251				BV_{CEO}	150	---	Vdc
3026	Emitter-base breakdown voltage	Bias Cond. D I_E=0.1mAdc			BV_{EBO}	10	---	Vdc
	Subgroup 3		5	3				
3076	Forward-current transfer ratio	Pulsed; 2/ I_C=20Adc V_{CE}=5Vdc			h_{FE}	30	120	---
3076	Forward-current transfer ratio	Pulsed; 2/ I_C=40Adc V_{CE}=5Vdc			h_{FE}	15	50	---
3076	Forward-current transfer ratio	Pulsed; 2/ I_C=70Adc V_{CE}=5Vdc			h_{FE}	10	40	---

Source: https://assist.dla.mil -- Downloaded: 20260-18T15:36Z
Check the source to verify that this is the current version before use.

Table I.- Group A inspection-(cont'd).

Test Method per MIL-STD-750	Examination or test 1/	Conditions	LTPD Non-TX	LTPD TX	Symbol	Limits Min	Limits Max	Unit
3076	Forward-current transfer ratio	Pulsed; 2/ I_C=90Adc V_{CE}=5Vdc			h_{FE}	5	---	---
3066	Base-emitter saturation voltage	Test Cond. A I_C=40Adc I_B=4Adc Pulsed; 2/			$V_{BE(sat)}$	---	1.8	Vdc
3066	Base-emitter saturation voltage	Test Cond. A I_C=70Adc I_B=10Adc Pulsed; 2/			$V_{BE(sat)}$	---	2.5	Vdc
3071	Collector-emitter saturation voltage:	I_C=40Adc I_B=4Adc Pulsed; 2/						
	2N4865				$V_{CE(sat)}$	---	1.0	Vdc
	2N5250				$V_{CE(sat)}$	---	1.0	Vdc
	2N5251				$V_{CE(sat)}$	---	1.35	Vdc
3071	Collector-emitter saturation voltage:	I_C=70Adc I_B=10Adc Pulsed; 2/						
	2N4865				$V_{CE(sat)}$	---	2.0	Vdc
	2N5250				$V_{CE(sat)}$	---	2.0	Vdc
	2N5251				$V_{CE(sat)}$	---	2.3	Vdc

Source: https://assist.dla.mil -- Downloaded: 202610-18T15:36Z
Check the source to verify that this is the current version before use.

Test Method per MIL-STD-750	Examination or test 1/	Conditions	LTPD Non-TX	LTPD TX	Symbol	Limits Min	Limits Max	Unit
	Subgroup 4		10	5				
3306	Magnitude of common-emitter small-signal short-circuit transfer ratio	I_C=5Adc V_{CE}=10Vdc f=10MHz			h_{fe}	1.0	7.0	---
3251	Pulse response:	Test Cond.A except test circuit and pulse requirements per Fig. 3A herein (I_C=70Adc)						
	Turn-on time	---			t_{on}	---	2.0	usec
	Storage time	---			t_s	---	1.5	usec
	Fall time	---			t_f	---	0.5	usec
	Subgroup 5		10	5				
3/	High-temperature operation:	T_A=+150°C 4/						
3041	Collector-emitter cutoff current	Bias Cond.C V_{CE}=100Vdc V_{BE}=0			I_{CES}	---	0.5	mAdc

Source: https://assist.dla.mil -- Downloaded: 202640-18T15:36Z
Check the source to verify that this is the current version before use.

Test Method per MIL-STD-750	Examination or test 1/	Conditions	LTPD Non-TX	LTPD TX	Symbol	Limits Min	Limits Max	Unit
---	Low-temperature operation:	T_A=-55°C 4/						
3076	Forward-current transfer ratio	Pulsed; 2/ I_C=40Adc V_{CE}=5Vdc			h_{FE}	10	---	---
3076	Forward-current transfer ratio	Pulsed; 2/ I_C=70Adc V_{CE}=5Vdc			h_{FE}	8	---	---

1/ See 3.3 herein.
2/ See 4.3.1 herein.
3/ In this Subgroup, the sample units subjected to the High-Temperature Operation test shall be permitted to return to and be stabilized at room ambient temperature prior to their being subjected to the Low-Temperature Operation test.

4/ Measurement(s) shall be made after thermal equilibrium has been reached at the temperature specified.

Source: https://assist.dla.mil -- Downloaded: 202610-18T15:36Z
Check the source to verify that this is the current version before use.

Test Method per MIL-STD-750	Examination or test 1/	Conditions	LTPD Non-TX	LTPD TX	Symbol	Limits Min	Limits Max	Unit
	Subgroup 1		20	20	---	---	---	---
2066	Physical dimensions	2/						
	Subgroup 2		15	15				
2031	Soldering heat	One cycle			---	---	---	---
1051	Temperature cycling	Test Cond.C			---	---	---	---
1056	Thermal shock (glass strain)	Test Cond.B			---	---	---	---
1021	Moisture resistance	No initial conditioning			---	---	---	---
	End-Point tests:							
3061	Emitter-base cutoff current	Bias Cond.D V_{EB}=8Vdc			I_{EBO}	---	5.0	uAdc
3041	Collector-emitter cutoff current	Bias Cond.C V_{CE}=100Vdc V_{BE}=0			I_{CES}	---	0.1	mAdc
3076	Forward-current transfer ratio	Pulsed; 3/ I_C=40Adc V_{CE}=5Vdc			h_{FE}	15	50	---

Source: https://assist.dla.mil -- Downloaded: 202610-18T15:36Z
Check the source to verify that this is the current version before use.

Table II.- Group B inspection.-(Cont'd).

Test Method per MIL-STD-750	Examination or test 1/	Conditions	LTPD Non-TX	LTPD TX	Symbol	Limits Min	Limits Max	Unit
	Subgroup 3		15	15				
2016	Shock	Non-operating; 1500G; 5 blows of 0.5 msec ea. in orientations X1, Y1, Y2, Z1 (total=20 blows)			---	---	---	---
2056	Vibration, variable frequency	---			---	---	---	---
2046	Vibration fatigue	Non-operating			---	---	---	---
2006	Constant acceleration (centrifugal)	5,000G Orientations X1, Y1, Y2, Z1			---	---	---	---
	End-Point tests: Same as for Subgroup 2 above							

Source: https://assist.dla.mil -- Downloaded: 2026-10-18T15:36Z
Check the source to verify that this is the current version before use.

Table II.- Group B inspection-(cont'd).

Test Method per MIL-STD-750	Examination or test 1/	Conditions	LTPD Non-TX	LTPD TX	Symbol	Limits Min	Limits Max	Unit
	Subgroup 4		20	20				
2036	Terminal strength, terminal torque	Test Cond. D1 Torque = 24 oz-in; t=15 sec., Base & emitter only			---	---	---	---
2036	Terminal strength, tension	Test Cond. A t=15 sec., ea. term., weight=10 lbs, (all terminals)			---	---	---	---
2036	Terminal strength, stud torque:	Test Cond. D2 Torque=120 lbs.-in., t=4/			---	---	---	---
	Measurements during stud-torque test:	End-Point tests as listed under sub-group 2 above			---	---	---	---
5/	Seal	Test Cond. C, Procedure IIIa; Test Cond. A for gross leaks			---	---	5×10^{-7}	atm cc/sec

Source: https://assist.dla.mil -- Downloaded: 202610-18T15:36Z
Check the source to verify that this is the current version before use.

Table II.- Group B inspection-(cont'd).

Test Method per MIL-STD-750	Examination or test 1/	Conditions	LTPD Non-TX	LTPD TX	Symbol	Limits Min	Limits Max	Unit
	Subgroup 5		20	20				
1041	Salt atmosphere (corrosion)	6/			---	---	---	---
	Subgroup 6		10	10				
---	Safe operating area (dc operation):	Test circuit and procedure per Fig. 4 herein 7/ T_C=+100°C t=60 sec. $t_r \leq$ 6.0 sec. $t_f \leq$ 6.0 Sec. 1 cycle			---	---	---	---
	Test #1: (all types)	I_C=50 Adc V_{CE}=4.0 Vdc (P_T = 200W)						
	Test #2: 2N5250: (only)	(P_T=200W) I_C=7.8 Adc V_{CE} = 26 Vdc						
	2N5251: (only)	I_C =5.4 Adc V_{CE} = 37 Vdc						

Source: https://assist.dla.mil -- Downloaded: 20260-18T15:36Z
Check the source to verify that this is the current version before use.

Table II.- Group B inspection-(cont'd).

Test Method per MIL-STD-750	Examination or test 1/	Conditions	LTPD Non-TX	LTPD TX	Symbol	Limits Min	Limits Max	Unit
	Subgroup 6-(cont'd)							
	Test #3:	T_C=+150°C						
		P_T=100W						
	2N4865:	V_{CE}=32 Vdc						
	2N5250:	V_{CE}=40 Vdc						
	2N5251:	V_{CE}=60 Vdc						
	Test #4:	T_C= +195°C						
		P_T= 10.0W						
	2N4865:	V_{CE}= 80 Vdc						
	2N5250:	V_{CE}= 100 Vdc						
	2N5251:	V_{CE}= 150 Vdc						
	End-point tests:							
	Same as for Subgroup 2 above							

Source: https://assist.dla.mil -- Downloaded: 202610-18T15:36Z
Check the source to verify that this is the current version before use.

Table II.- Group B inspection-(cont'd).

Test Method per MIL-STD-750	Examination or test 1/	Conditions	LTPD Non-TX	LTPD TX	Symbol	Limits Min	Limits Max	Unit
	Subgroup 7		10	10				
8/	Unclamped inductive sweep	T_A= +25°C I_B=500 mAdc I_C = 5Adc			---	---	---	---
	End-Point Tests: Same as for Subgroup 2 above							
	Subgroup 8		10	10				
9/	Clamped inductive sweep	T_A=+25°C I_B= 3Adc I_C=30 Adc			---	---	---	---
	End-Point Tests: Same as for Subgroup 2 above							

Source: https://assist.dla.mil -- Downloaded: 202610-18T15:36Z
Check the source to verify that this is the current version before use.

Table II.- Group B inspection-(cont'd).

Test Method per MIL-STD-750	Examination or test 1/	Conditions	LTPD Non-TX	LTPD TX	Symbol	Limits Min	Limits Max	Unit
	Subgroup 9		10	λ=5				
1031	High-temperature life (non-operating)	T_{stg}=+200°C t=340 hrs 10/			---	---	---	---
	End-Point Tests:							
3061	Emitter-base cutoff current	Bias Cond.D V_{EB}= 8Vdc			I_{EBO}	---	7.0	uAdc
3041	Collector-emitter cutoff current	Bias Cond.C V_{CE}= 100 Vdc V_{BE} = 0			ΔI_{CES}	---	11/	12/
3076	Forward-current transfer ratio	Pulsed; 3/ I_C= 40 Adc V_{CE} = 5 Vdc			Δh_{FE}	---	13/	12/

Source: https://assist.dla.mil -- Downloaded: 202610-18T15:36Z
Check the source to verify that this is the current version before use.

Table II.- Group B inspection-(cont'd).

Test Method per MIL-STD-750	Examination or test 1/	Conditions	LTPD Non-TX	LTPD TX	Symbol	Limits Min	Limits Max	Unit
	Subgroup 10		10		λ=5			
1026	Steady state operation life:	T_C=+100°C t= 340 hrs 10/ P_T= 200 W V_{CE}=20 Vdc			---	---	---	---
	End-Point Tests: Same as for Sub-group 9 above							

1/ See 3.3 herein. 2/ See 4.3.3 herein. 3/ See 4.3.1 herein.

4/ With the transistor mounted in normal mounting position, the specified torque shall be maintained (torque-wrench removed) throughout each of the specified End-Point tests and measurements.

5/ Per Method 112 in Standard MIL-STD-202.

6/ Electrical rejects from the same lot under evaluation may be used for this test.

7/ See 4.3.4 herein. 8/ See Fig. 6 herein. 9/ See Fig. 7 herein.

10/ See 4.2.3 herein. 11/ 100% or 20uA, whichever is greater. 12/ Change from initial Group A reading.

13/ ±20%; however, h_{FE} ≥ 15 shall be met.

Source: https://assist.dla.mil -- Downloaded: 202610-18T15:36Z
Check the source to verify that this is the currentversion before use.

Table III.- Group C inspection. 1/

Test Method per MIL-STD-750	Examination or test 2/	Conditions	LTPD Non-TX	LTPD TX	Symbol	Limits Min	Limits Max	Unit
	Subgroup 1		20	20				
3151	Thermal resistance (junction-to-case)	---			θ_{J-C}	---	0.5	°C/W
3251	Pulse response:	Test Cond. A except test circuit and pulse reqts. per Fig. 3B herein. (I_C=40 Adc)						
	Turn-on time	---			t_{on}	---	2.0	usec
	Storage time	---			t_s	---	1.5	usec
	Fall time	---			t_f	---	0.5	usec
	Subgroup 2		λ=10	---				
1031	High-temperature life(non-operating):	T_{stg}=+200°C t= 1000 hrs 3/			---	---	---	---

Source: https://assist.dla.mil -- Downloaded: 20261O-18T15:36Z
Check the source to verify that this is the current version before use.

Table III.- Group C inspection. 1/-(cont'd).

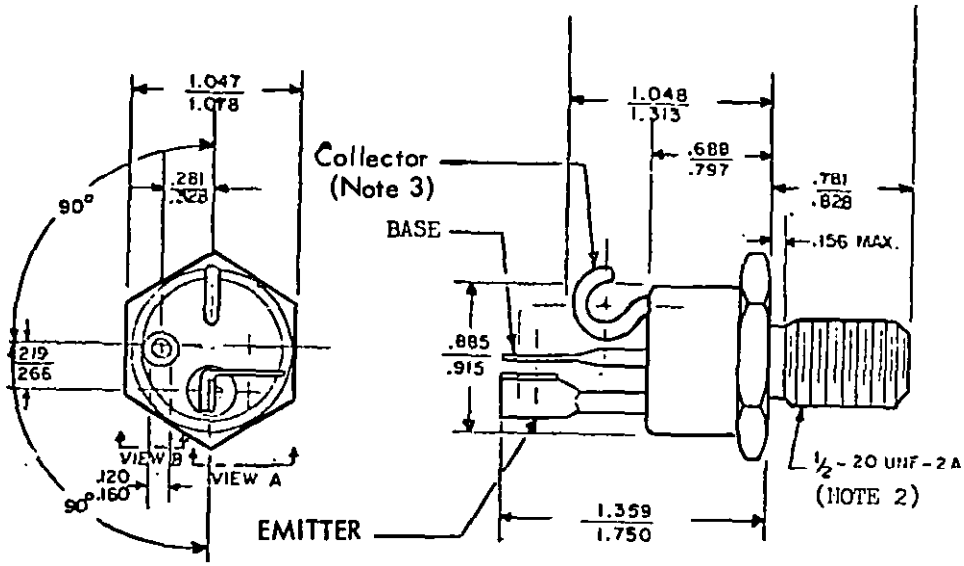
Test Method per MIL-STD-750	Examination or test 2/	Conditions	LTPD Non-TX	LTPD TX	Symbol	Limits Min	Limits Max	Unit
	Subgroup 2-(cont'd): End-Point Tests:							
3061	Emitter-base cutoff current	Bias Cond.D V_{EB}=8 Vdc			I_{EBO}	---	7.0	uAdc
3041	Collector-emitter cutoff current	Bias Cond.C V_{CE}=100 Vdc V_{BE} = 0			ΔI_{CES}	---	4/	5/
3076	Forward-current transfer ratio	Pulsed; 6/ I_C=40 Adc V_{CE} =5 Vdc			Δh_{FE}	---	7/	5/
	Subgroup 3		λ=10	---				
1026	Steady state operation life:	T_C=+100°C t=1000 hrs V_{CE}=20 Vdc P_T = 200W 3/			---	---	---	---
	End-Point Tests: Same as for Subgroup 2 above							

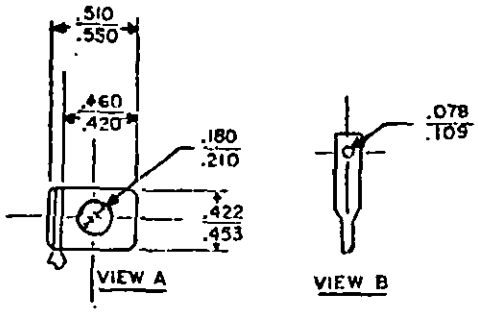
1/ See 4.2.4 herein. 2/ See 3.3 herein. 3/ See 4.2.3 herein.

4/ 100% or 20uA, whichever is greater. 5/ Change from initial Group A reading.

6/ See 4.3.1 herein. 7/ ±20%; however, $h_{FE} \geq 15$ shall be met.

Source: https://assist.dla.mil -- Downloaded: 202610-18T15:36Z
Check the source to verify that this is the current version before use.



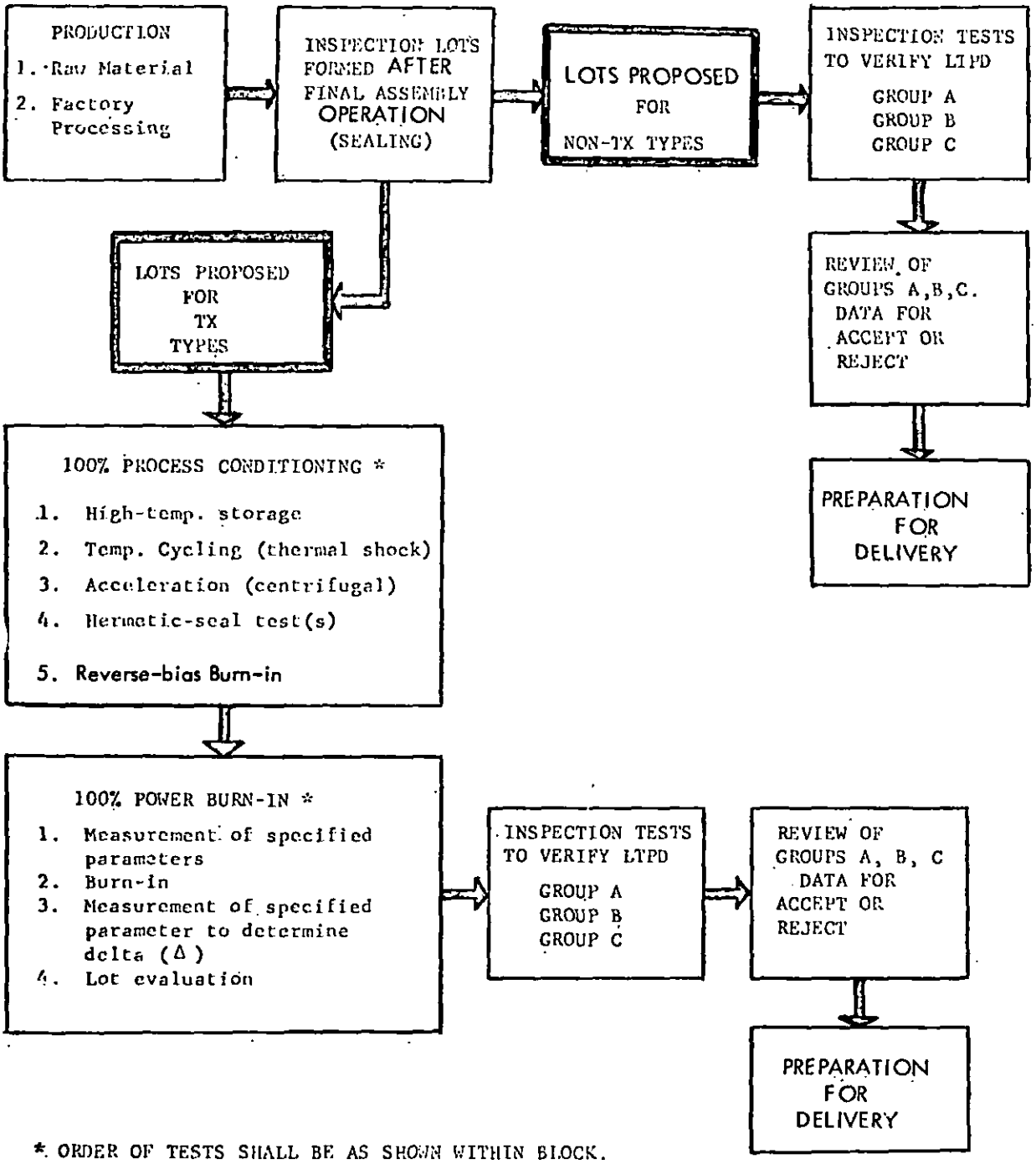


NOTES:

1. All dimensions in inches, unless otherwise specified.
2. Maximum PITCH DIA of plated thread shall be basic P.D.(.4675).
3. Collector is electrically connected to case.

Figure 1. Outline and dimensions.

Source: https://assist.dla.mil -- Downloaded: 2026-10-18T15:36Z
Check the source to verify that this is the current version before use.



* ORDER OF TESTS SHALL BE AS SHOWN WITHIN BLOCK.

FIGURE 2. FLOW CHART OF NON-TX AND TX TYPES.

Source: https://assist.dla.mil -- Downloaded: 202610-18T15:36Z
Check the source to verify that this is the current version before use.

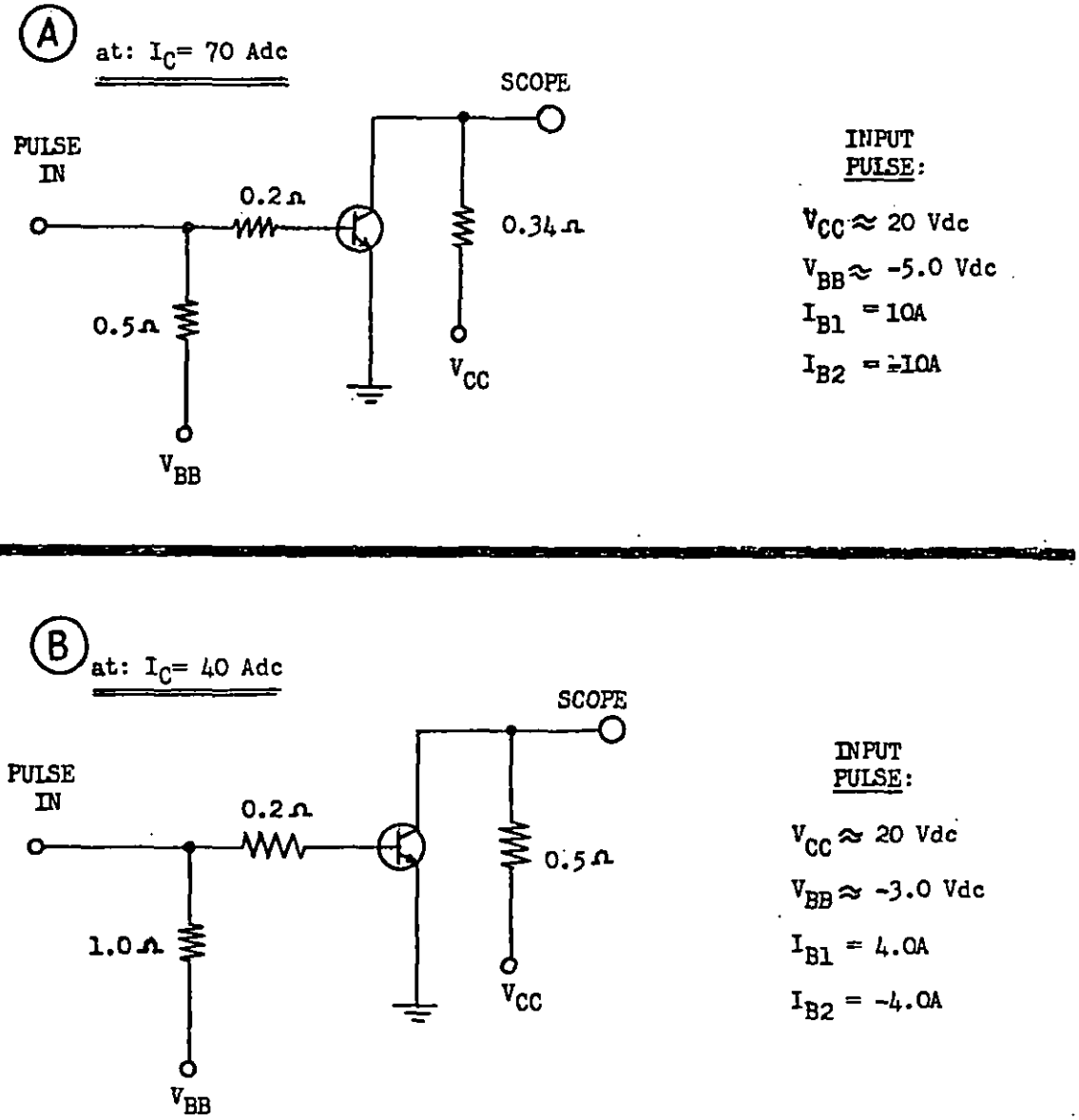
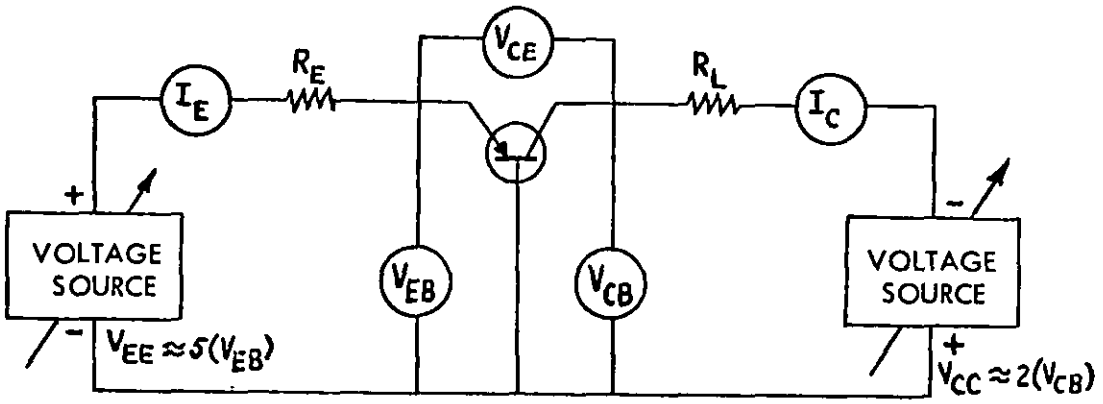


Fig. 3. Pulse Response test circuit(s).

Source: https://assist.dla.mil -- Downloaded: 202610-18T15:36Z
Check the source to verify that this is the current version before use.



TEST PROCEDURE:

a. Starting with V_{CC} and V_{EE} at a low value, increase V_{CC} to obtain approximately the specified V_{CE}; increase V_{EE} to obtain approximately the specified I_C. Increase V_{CC} and subsequently adjust V_{EE} to obtain the specified V_{CE} and I_C. Operate the transistor at specified case temp; for the specified time duration.

b. At end of operation time: decrease V_{CC} to obtain V_{CE} near zero; turn off V_{EE}; turn off V_{CC}; remove transistor-under-test.

Fig. 4. Safe Operating Area (DC Operation) test circuit.

Source: https://assist.dla.mil -- Downloaded: 2026-10-18T15:36Z
Check the source to verify that this is the current version before use.

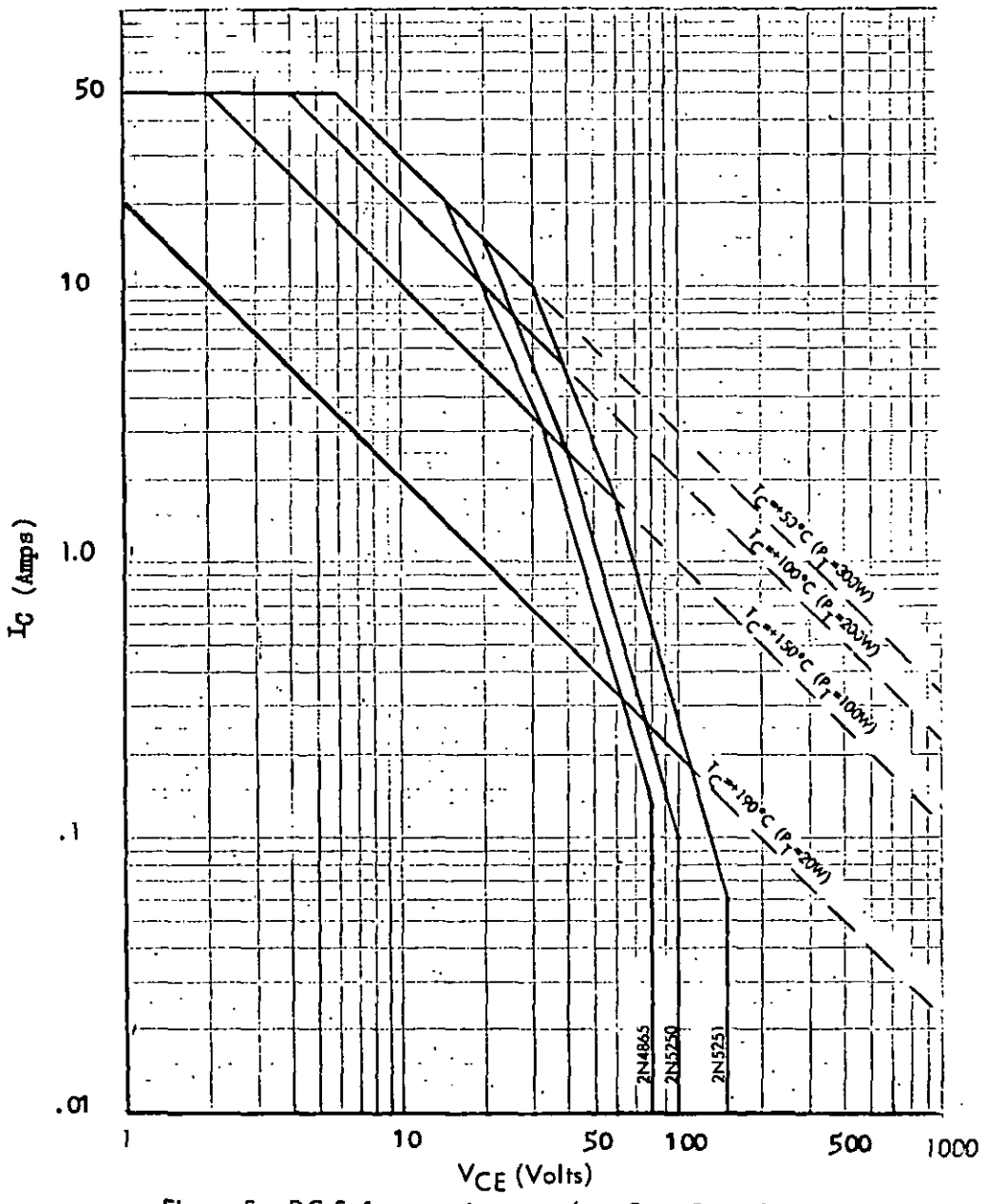
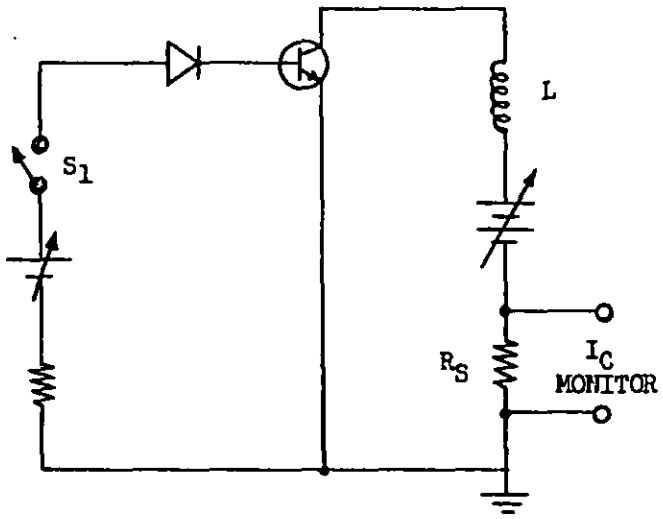


Figure 5. DC Safe operating area (per Case Temp.) nomograph.

Source: https://assist.dla.mil -- Downloaded: 20261O-18T15:36Z
Check the source to verify that this is the current version before use.



TESTING PROCEDURE:

1. With switch S_1 closed, apply specified I_B and I_C (at specified T_C cond.)
2. As soon as 1 above is effected and ascertained, open switch S_1. Device fails if, now, second breakdown occurs.
3. Perform specified end-point tests.

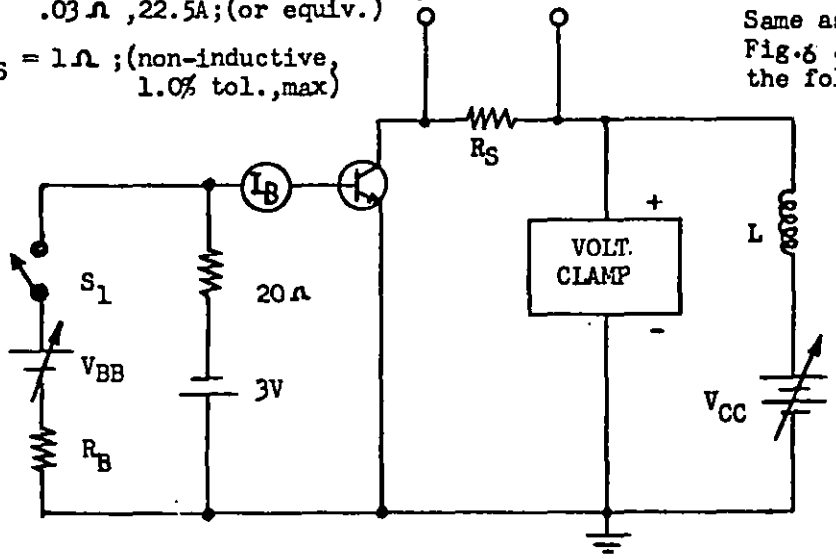
L = 5 mH; (see Fig.5 data)

R_S = 1 Ω ; (non-inductive, 1.0%-tol.,max.)

Fig. 6. Unclamped Inductive Sweep Test circuit.

L = 5 mH; Chicago St'd. Transf. Corp. Type C=2689 Filter Choke, .03 Ω ,22.5A;(or equiv.)

R_S = 1 Ω ;(non-inductive, 1.0% tol.,max)



TESTING PROCEDURE:

Same as specified in Fig.6 above, except use the following respective Clamp voltages:

2N4865 = 100 $^{+0}_{-5}$ V

2N5250 =125 $^{+0}_{-5}$ V

2N5251 =180 $^{+0}_{-5}$ V

Special note:

Device fails if clamp voltage is not reached.

Fig.7 . Clamped Inductive Sweep Test circuit.

Source: https://assist.dla.mil -- Downloaded: 202610-18T15:36Z
Check the source to verify that this is the current version before use.

5. PREPARATION FOR DELIVERY

5.1 Preparation for delivery.- Preparation for delivery, and the inspection of preparation for delivery, shall be in accordance with Specification MIL-S-19500.

6. NOTES

6.1 Notes.- The notes included in Specification MIL-S-19500, with the following additions or exceptions, are applicable to this document.

6.2 Application guidance.- To insure proper circuit application, particular attention should be given to the differential voltage and gain characteristics, and ratings, pertinent to the individual transistor types covered herein.

6,3 Ordering data.- Lot-inspection data: See 4.2.6 herein.

6.4 Procurement of "TX" types.- See 4.2.2 herein.

6.5 Quality assurance for "TX" types.- It is expected that where the permanent lot record, maintained as required in 4.4.1 herein, begins to show a condition of failures in excess of the LTPD rate specified herein, or a condition of entire-lot rejection (see 4.4.7.4 herein) is recorded, the Government procuring activity will determine any additional monitoring controls and evaluation criteria needed to attain restoration of the prescribed (herein) quality assurance for acceptance of product.

6.6 Qualification.- With respect to products requiring qualification, awards will be made only for such products as have, prior to the time set for opening of bids, been tested and approved for inclusion in Qualified Products List (QPL)-19500, whether or not such products have actually been so listed by that date. Information pertaining to qualification of products covered by this specification should be requested from the Commanding General, US Army Electronics Command, Fort Monmouth, New Jersey, Attention: AMSEL-PP-EM-2.

CUSTODIAN:

ARMY-EL

Preparing activity:

ARMY-EL

Project No. 5961 - A183

Source: https://assist.dla.mil -- Downloaded: 202640-18T15:36Z
Check the source to verify that this is the currentversion before use.

STANDARDIZATION DOCUMENT IMPROVEMENT PROPOSAL

(See Instructions – Reverse Side)

1. DOCUMENT NUMBER	2. DOCUMENT TITLE

3a. NAME OF SUBMITTING ORGANIZATION	4. TYPE OF ORGANIZATION *(Mark one)*
	☐ VENDOR
b. ADDRESS *(Street, City, State, ZIP Code)*	☐ USER
	☐ MANUFACTURER
	☐ OTHER *(Specify)*:

5. PROBLEM AREAS

a. Paragraph Number and Wording:

b. Recommended Wording:

c. Reason/Rationale for Recommendation:

6. REMARKS

7a. NAME OF SUBMITTER *(Last, First, MI)* – Optional	b. WORK TELEPHONE NUMBER *(Include Area Code)* – Optional
c. MAILING ADDRESS *(Street, City, State, ZIP Code)* – Optional	8. DATE OF SUBMISSION *(YYMMDD)*

(TO DETACH THIS FORM, CUT ALONG THIS LINE.)

DD FORM 82 MAR 1426 PREVIOUS EDITION IS OBSOLETE.

Source: https://assist.dla.mil -- Downloaded: 2026-10-18T15:36Z
Check the source to verify that this is the current version before use.